

RACINE BATTEN, WI, USA

WMO: 726424

Lat: 42.761N

Lon: 87.814W

Elev: 205

StdP: 98.88

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 94818

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-16.4	-24.7	0.4	-18.3	-22.4	0.5	-15.7	10.7	-3.5	9.6	-2.6	4.2	280	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	31.5	23.3	29.7	22.5	28.0	21.5	24.7	29.7	23.6	28.1	22.7	26.8	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.9	18.1	27.6	22.3	17.4	26.7	21.3	16.4	25.5	75.6	29.9	71.5	28.1	67.9	26.8	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.3	8.9	8.1	DB	-22.0	34.3	3.3	2.2	-24.4	35.9	-26.3	37.1	-28.2	38.4	-30.6	39.9
(5)				WB	-22.5	26.1	3.2	1.2	-24.8	27.0	-26.7	27.6	-28.5	28.3	-30.8	29.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-4.8	-3.8	1.8	7.5	13.2	18.9	22.3	21.6	17.9	11.1	4.5	-2.0
	DBStd	10.68	6.22	5.84	5.55	4.62	4.67	4.14	3.38	2.88	4.11	4.68	5.26	5.87
	HDD10.0	1825	460	388	263	103	19	0	0	1	43	177	371	
	HDD18.3	3733	718	621	514	327	171	42	6	58	228	415	629	
	CDD10.0	1492	0	0	7	28	119	268	382	359	237	78	12	1
	CDD18.3	358	0	0	0	1	13	61	129	106	43	4	0	0
	CDH23.3	2557	0	0	1	25	105	467	1029	651	256	24	0	0
	CDH26.7	709	0	0	0	4	19	129	333	164	57	3	0	0
Wind	WSAvg	4.0	4.4	4.4	4.4	4.8	4.1	3.5	3.4	3.3	3.6	4.1	4.3	4.3
Precipitation	PrecAvg	853	43	34	60	92	85	93	89	91	92	61	69	47
	PrecMax	1205	119	95	192	199	344	226	207	213	229	182	169	132
	PrecMin	575	4	0	9	23	12	20	36	10	1	4	15	5
	PrecStd	139	28	22	36	41	61	49	38	47	59	38	38	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	13.2	21.0	27.1	28.8	32.4	35.3	32.8	30.7	26.3	19.5	13.9
		MCWB	9.4	9.0	15.5	17.0	20.0	23.1	25.0	24.2	22.6	19.5	14.6	11.0
	2%	DB	7.5	9.1	16.3	22.2	26.2	30.1	32.2	30.3	28.1	22.8	16.8	10.0
		MCWB	4.8	5.9	12.7	15.1	18.7	21.9	24.2	23.1	21.3	17.7	12.1	7.7
	5%	DB	4.8	6.0	12.7	18.2	23.8	27.8	30.0	28.3	26.3	20.9	14.1	7.4
		MCWB	3.1	3.4	8.7	12.7	17.6	21.0	23.0	22.0	20.3	15.9	10.8	5.9
	10%	DB	3.1	3.5	9.4	15.2	21.6	26.2	28.2	26.9	24.1	18.3	12.2	5.8
		MCWB	1.9	1.8	6.2	10.2	16.1	19.9	22.0	21.1	19.5	14.6	9.2	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	10.0	15.7	18.8	21.7	25.0	26.7	25.4	24.0	21.0	15.3	12.2
		MCDB	11.3	12.8	19.7	24.8	27.0	29.8	32.4	31.3	28.7	25.1	18.1	13.8
	2%	WB	5.4	5.8	12.9	15.7	19.8	23.2	25.1	24.1	22.5	18.8	13.1	8.0
		MCDB	6.9	8.6	16.0	21.3	24.2	28.4	30.5	28.3	26.5	21.2	15.8	8.9
	5%	WB	3.3	3.5	9.0	13.3	18.5	22.0	24.0	23.1	21.5	16.9	11.2	6.1
		MCDB	4.6	5.6	11.9	18.1	22.8	26.9	28.6	27.0	24.9	19.5	13.7	7.4
	10%	WB	1.9	1.9	6.4	10.6	16.9	20.7	23.0	22.2	20.3	15.2	9.3	4.0
		MCDB	3.3	3.4	9.0	14.4	20.8	24.8	27.2	25.6	23.6	18.2	12.0	5.5
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.5	7.7	8.6	9.3	9.3	8.9	8.2	8.8	8.3	7.4	6.8
		MCDBR	8.5	9.8	12.1	14.5	13.0	11.3	10.6	10.1	10.7	11.3	10.2	8.2
		MCWBR	7.2	7.4	8.9	9.1	7.5	6.0	5.1	4.8	5.4	6.9	7.5	7.2
	5% WB	MCDBR	8.0	9.3	11.6	13.9	12.0	10.5	9.7	8.7	9.4	9.6	9.3	7.6
		MCWBR	7.1	7.4	9.2	9.4	7.7	6.3	5.3	4.7	5.3	7.2	7.5	7.1
Clear-Sky Solar Irradiance	taub	0.292	0.311	0.343	0.391	0.416	0.436	0.432	0.424	0.395	0.349	0.312	0.291	
	taud	2.447	2.393	2.394	2.292	2.275	2.270	2.280	2.301	2.357	2.480	2.524	2.490	
	Ebn at Noon	852	890	900	875	860	841	839	834	833	836	825	823	
	Edh at Noon	74	93	105	126	132	133	130	123	109	85	69	66	
All-Sky Solar Radiation	RadAvg	1.46	2.35	3.55	4.58	5.30	6.04	6.14	5.30	4.29	2.76	1.72	1.20	
	RadStd	0.19	0.28	0.31	0.39	0.42	0.40	0.28	0.33	0.36	0.31	0.17	0.13	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page